

JNTU Hyderabad

REGULAR EXAMINATION ? 2026

Design of Steel Structures ? Regular Examination, 2026

Subject Code: CE501 Semester: 5 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Bolted and Welded Connections)

1. List the advantages and limitations of Bolted and Welded Connections. Explain any two in detail. [12 Marks]
2. How is Bolted and Welded Connections implemented? Give an algorithm / procedure and explain. [12 Marks]
3. Define Bolted and Welded Connections. Explain its significance in Design of Steel Structures with a suitable example. [8 Marks]

SECTION B (Unit 2: Tension Members)

4. List the advantages and limitations of Tension Members. Explain any two in detail. [12 Marks]
5. How is Tension Members implemented? Give an algorithm / procedure and explain. [12 Marks]
6. Write short notes on Tension Members. [8 Marks]

SECTION C (Unit 3: Compression Members)

7. List the advantages and limitations of Compression Members. Explain any two in detail. [8 Marks]
8. Solve a problem related to Compression Members. Show all the steps clearly. [12 Marks]
9. Define Compression Members. Explain its significance in Design of Steel Structures with a suitable example. [8 Marks]

SECTION D (Unit 4: Beams and Plate Girders)

10. How is Beams and Plate Girders implemented? Give an algorithm / procedure and explain. [12 Marks]
11. Define Beams and Plate Girders. Explain its significance in Design of Steel Structures with a suitable example. [12 Marks]
12. Solve a problem related to Beams and Plate Girders. Show all the steps clearly. [12 Marks]

SECTION E (Unit 5: Plastic Design)

13. Derive the expression related to Plastic Design and explain each term. [12 Marks]

14. Discuss the engineering applications of Plastic Design. [12 Marks]

15. Solve a problem related to Plastic Design. Show all the steps clearly. [12 Marks]

*** End of Question Paper ***